1 Let S be a stack of size $n \geq 1$. Starting with the empty stack, suppose we push the first n natural numbers in sequence, and then perform n pop operations. Assume that Push and Pop operation take X seconds each , and Y seconds elapse between the end of the one such stack operation and the start of the next operation. For $m \geq 1$, define the stack-life of m as the time elapsed from the end or Push (m) to the start of the pop operation that removes m from S . The average stack-life of an element of this stack is

A n(X+Y)

B 3Y+2X

C n(X+Y)-X

D Y+2X

2 A circular queue has been implemented using an array of size 100. If front = 8 and rear = 3, how many elements are currently in the queue?

3

What does fun(78) print ?

```python
from collections import deque
#S is stack with list implementation
def fun(n):
    q = deque()
    s = []

    i = n
    while i > 0:
        remainder = i % 4
        q.append(remainder)
        i = i // 4

    while q:
        a = q.popleft() #deque_operation
        s.append(a)

    while s:
        print(s.pop(), end='')
```

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4

Consider a sorted circular doubly-linked list where the head element points to the smallest element in the list. What is the tightest complexity of finding the median element in the list?

A O(1)

B O(logn)

C O(nlogn)

D O(n)

5

Following is Python like pseudo code of a function that takes a number as an argument, and uses a queue Q and stack S to do processing.
What does this fun(n) function do !

```
from collections import deque
#S is stack with list implementation
def fun(n):
    q = deque()
    s = []

    i = n
    while i > 0:
        remainder = i % 8
        q.append(remainder)
        i = i // 8

    while q:
        a = q.popleft() #deque_operation
        s.append(a)

    while s:
        print(s.pop(), end='')
```

6

A queue can be simulated using two stacks. If n is the number of elements currently in the queue, while an dequeue operation takes 1 pop, a enqueue operation needs:

A 2n+1 pushes and 2n pops

B 2n pushes and 2n pops

C 2n+1 pushes and n pops

D n+1 pushes and 2n pops

7

What is the prefix expression for the infix expression ((a+b*(c-d)-e^f)+(g/h))?

A + + a * b - - c d ^ e f / g h

B + + a - - * b c d ^ e f / g h

C + + a - * b - c d ^ e f / g h

D + - + a * b - c d ^ e f / g h

8

What does this code do !

```
def do_something(n):
    s = []
    s.append(0)
    s.append(1)

    for i in range(n):
        a = s.pop()
        b = s.pop()
        s.append(a)
        s.append(a + b)
        print(b)
```

A Print first n fibonacci numbers

B Print numbers from n to 0

C Print numbers from 1 to n

9 Consider a sequence of push and pop operations used to push the integers 0 through 9 on the stack. The numbers will be pushed in order; however, the pop operations can be interleaved with the push operations, and can occur any time. We will perform pop when there is at least one item in the stack. When an item is popped, it is printed to the terminal. Which of the following could be the output from such a sequence of operations?

A 0123456789

B 4321056789

C 5678901234

D 4321098765

10 The prefix expression `* - + 4 3 5 / + 2 4 3` Evaluates to

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11

If the expression `((2 + 3) * 4 + 5 * (6 + 7) * 8) + 9` is evaluated with * having precedence over +, then the value obtained is same as the value of which of the following prefix expressions?

A `+ * + + 2 3 4 * * 5 + 6 7 8 9`

B `* + + + 2 3 4 * * 5 + 6 7 8 9`

C `+ + * + 2 3 4 * * 5 + 6 7 8 9`

D `+ + + * 2 3 4 * * 5 + 6 7 8 9`

12

A circular queue is implemented with the help of an array and currently with data items in `data[5]` to `data[9]`. The capacity of the array is 10.
The enqueue operation place's the new entry in the array at data[x] what is teh value of x

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13 Suppose that we use a linked list to represent a queue and that in addition to the enqueue and dequeue functions add a new operation to the queue that deletes the last element of the queue. Which of the following linked structures do we need to use to guarantee that this operation is also executed in constant time?

A Singly linked list with both front and rear pointer.

B Singly circular linked list with both front and rear pointer.

C Singly circular linked list with rear pointer only.

D Doubly linked list with both front and rear pointer

14

Following is a Python pseudocode of a function which takes a queue as an argument and a stack S to to the rest processing :

```
#Assume Q is a queue with n elements
def foo(Q):
    S = Stack()

    while not Q.is_empty():
        item = Q.dequeue()
        S.push(item)

    while not S.is_empty():
        item = S.pop()
        Q.enqueue(item)
```

(A) Reverse the Q

(B) Makes the Q empty

15

Suppose that a queue q contains the value 10, 30, 20, 60, 50, 40 in that order. with 10 at the front of q. Suppose that there are just three operations that can be performed using only one stack, s.

I. Dequeue x from q then print x

II. Dequeue x from q then s. push(x);

III. Pop x from s then print x.

Which of the following is a possible output using just these operations?

A 10,30,20,60,50,40

B 20,30,10,40,50,60

C 30,10,60,20,40,50

D 10,50,40,30,20,60

16

A queue contains the elements A, B, C, D, E with A being at the front of the queue. What is the minimum number of dequeue and enqueue operations needed to generate a queue containing D, E, C, A, B (in that order, with D being at the front of the queue, with the help of an empty queue)?

A 2 dequeues and 2 enqueues

B 3 dequeues and 3 enqueues

C 5 dequeues and 5 enqueues

D 2 dequeues and 3 enqueues

17

Consider an infix expression: 4 + 3 * (6 * 3 - 12). Suppose that we are using the usual stack algorithm to convert the expression from infix to postfix notation. What is the maximum number of symbols that will appear on the stack AT ONE MIME during the conversion of this expression?

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